

The axioms of n -exact categories

BACHELOR'S THESIS

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Abstract

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Introduction

0.1 Prerequisites

In order to understand this paper, the reader should have some knowledge about modern homological algebra. In particular, they should already be familiar with category theory, more specifically abelian category theory, as is presented for example in the lecture notes of MA3204 Homological Algebra [\[3\]](#)

Conventions and notations

Chapter 1

An Introduction to Exact Categories

As a way to generalize the notion of abelian categories, one can look at a general additive category on which we define a certain notion of short exact sequences that has very similar properties to the abelian category counterpart. This chapter aims to motivate the definition of these so-called exact categories, to later define them formally and generalize some classical results from abelian categories.

In this chapter, the word kernel (respectively cokernel) will usually refer to the kernel morphism (respectively cokernel morphism), not the object. The content of this chapter is inspired by both the lecture notes of MA3204 Homological Algebra [3] and the article about exact categories by Theo Bühler [1].

1.1 Motivation

When studying abelian categories, there are many convenient properties related to monics and epics, which are used to prove results about short exact sequences. Let us recall a few:

Proposition 1.1 *In an abelian category,*

- *every monic is a kernel of its cokernel;*
- *every epic is a cokernel of its kernel.*

Furthermore, the following properties hold:

- *the pushout of a monic is a monic;*
- *the pullback of a monic is a monic;*
- *the pushout of an epic is an epic;*
- *the pullback of an epic is an epic.*

We refer the reader to [3, Sections 12-13] for the proofs of these facts.

Fundamentally, this tells us that every monic and every epic fits into a short exact sequence, and that taking the pullback or pushout of a morphism that is part of a short exact sequence gives rise to another short exact sequence. Indeed, if $A \xrightarrow{f} B$ is monic, then it fits into the short exact sequence $0 \rightarrow A \xrightarrow{f} B \xrightarrow{\text{cok}f} \text{Cok } f \rightarrow 0$. Furthermore, since morphisms that are part of a short exact sequence are either monic or epic, they will once again be monic or epic after a pullback or pushout, thus will once again fit in a short exact sequence.

These results allow us to conclude many properties about kernels and cokernels:

Proposition 1.2 *Let $\mathcal{A}$ be an abelian category and let $\mathcal{E}$ be the class of short exact sequences. $\mathcal{E}$ is closed under isomorphism. That is, if $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ is a short exact sequence and we can form the commutative diagram*

$$\begin{array}{ccccccc}
 0 & \longrightarrow & A & \xrightarrow{f} & B & \xrightarrow{g} & C \longrightarrow 0 \\
 & & \downarrow a & & \downarrow b & & \downarrow c \\
 0 & \longrightarrow & A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' \longrightarrow 0
 \end{array}$$

where the vertical arrows are isomorphisms, then $0 \rightarrow A' \xrightarrow{f'} B' \xrightarrow{g'} C' \rightarrow 0$ is a short exact sequence.

Furthermore, the following hold:

- [E0] For any object $A \in \mathcal{A}$, the identity morphism 1_A is a kernel;
- [E0^{op}] For any object $A \in \mathcal{A}$, the identity morphism 1_A is a cokernel;
- [E1] The class of kernels is closed under composition;
- [E1^{op}] The class of cokernels is closed under composition;
- [E2] The pushout of a kernel along any morphism is a kernel;
- [E2^{op}] The pullback of a cokernel along any morphism is a cokernel.

Proof It is clear that the class of short exact sequences is closed under isomorphism. Indeed, $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ is a short exact sequence if and only if f is a kernel of g and g is a cokernel of f . The result follows from the fact that kernels and cokernels are preserved by isomorphisms.

[E0] and [E0^{op}] follow directly from the fact that the identity morphism is both the kernel and cokernel of the zero map.

We can deduce [E1] and [E1^{op}] by recalling that composing monics (respectively epics) gives us monics and that monics are kernels of their cokernel (respectively epics are cokernels of their kernel).

As for [E2], recall that kernels are always monic. Thus, by 1.1, the pushout of a kernel is monic as well. But then, by 1.1 again, it is a kernel of its cokernel, thus it is a kernel. The proof for [E2^{op}] is similar.

□

Remark 1.4 We can in a very similar way prove that the pullback of a kernel (respectively the pushout of a cokernel) is a kernel (respectively cokernel), however we choose not to include this as it turns out to be unnecessary for many results.

1.2 Exact Categories

Based on the observations from the last section, one can define a more general notion that encapsulates the idea of short exact sequences, namely exact categories. To do so, let us first define kernel-cokernel pairs.

Definition 1.5 Let $\mathcal{A}$ be an additive category. A *kernel-cokernel pair* (i, p) in $\mathcal{A}$ is a pair of composable morphisms $A \xrightarrow{i} B \xrightarrow{p} C$ such that i is a kernel of p and p is a cokernel of i .

In the context of abelian categories, we notice that kernel-cokernel pairs indeed give rise to short exact sequences in the usual sense. In a general additive category, however, we cannot say the same since we have no guarantee that kernels and cokernels even exist for arbitrary morphisms, and thus we are unsure that $\text{Im } i$ exists. But we will use our intuition from abelian categories to expand the notion of short exact sequences to additive categories. First, let us recall what it means for a class to be closed under isomorphism.

Definition 1.6 Let $\mathcal{A}$ be an additive category and $\mathcal{E}$ be a class of kernel-cokernel pairs in $\mathcal{A}$. We call a morphism $A \xrightarrow{i} B$ an *admissible monic* (or *inflation*) if there exists a morphism $B \xrightarrow{p} C$ such that $(i, p) \in \mathcal{E}$. Dually, we call a morphism $B \xrightarrow{q} C$ an *admissible epic* (or *deflation*) if there exists a morphism $A \xrightarrow{j} B$ such that $(j, q) \in \mathcal{E}$. We use $A \xrightarrow{i} B$ for admissible monics, and $B \xrightarrow{p} C$ for admissible epics. We

say $\mathcal{E}$ is closed under isomorphism if whenever we have $A \xrightarrow{i} B \xrightarrow{p} C \in \mathcal{E}$ and the following commutative diagram

$$\begin{array}{ccccc} A & \xrightarrow{i} & B & \xrightarrow{p} & C \\ \downarrow a & & \downarrow b & & \downarrow c \\ A' & \xrightarrow{j} & B' & \xrightarrow{q} & C' \end{array}$$

with a , b , and c isomorphisms, then $(j, q) \in \mathcal{E}$.

It is usually convenient to require being closed under isomorphism, as objects with universal properties are usually unique up to isomorphism. In particular, recall that all kernel objects of a specific morphism are isomorphic. In the case of a class of kernel-cokernel pairs $\mathcal{E}$, being closed under isomorphism implies that all kernels of an admissible epic are admissible monics. Indeed, consider $B \xrightarrow{p} C$ an admissible epic and $A \xrightarrow{i} B$ an admissible monic such that $(i, p) \in \mathcal{E}$. Let $A' \xrightarrow{j} B$ be another kernel of p and $A \xrightarrow{a} A'$ be the isomorphism between both kernels. We can construct the following commutative diagram

$$\begin{array}{ccccc} A & \xrightarrow{i} & B & \xrightarrow{p} & C \\ \downarrow a & & \downarrow 1_B & & \downarrow 1_C \\ A' & \xrightarrow{j} & B & \xrightarrow{p} & C \end{array}$$

Since all vertical arrows are isomorphisms, we can conclude that $(j, p) \in \mathcal{E}$. Dually, we can show that all cokernels of an admissible monic are admissible epics.

We can now proceed to define exact categories, basing our definition on the properties we established for abelian categories:

Definition 1.7 Let $\mathcal{A}$ be an additive category. An *exact structure* on $\mathcal{A}$ is a class $\mathcal{E}$ of kernel-cokernel pairs in $\mathcal{A}$ that is closed under isomorphism and satisfies the following axioms:

- [E0] For all objects $A \in \mathcal{A}$, the identity morphism 1_A is an admissible monic;
- [E0^{op}] For all objects $A \in \mathcal{A}$, the identity morphism 1_A is an admissible epic;
- [E1] The composition of two admissible monics yields an admissible monic;
- [E1^{op}] The composition of two admissible epics yields an admissible epic;
- [E2] The pushout of an admissible monic along any morphism yields an admissible monic;
- [E2^{op}] The pullback of an admissible epic along any morphism yields an admissible epic.

The last two axioms can be summarized by the following commutative diagrams:

$$\begin{array}{ccc}
 C & \xrightarrow{f'} & D \\
 \downarrow c & \text{PO} & \downarrow d \\
 C' & \xrightarrow{\quad f \quad} & D'
 \end{array}
 \quad \text{and} \quad
 \begin{array}{ccc}
 C' & \overset{\quad f' \quad}{\dashrightarrow} & D' \\
 \downarrow c & \text{PB} & \downarrow d \\
 C & \xrightarrow{\quad f \quad} & D
 \end{array}$$

If $\mathcal{E}$ is an exact structure on $\mathcal{A}$, we say $(\mathcal{A}, \mathcal{E})$ is an *exact category* and we call the elements of $\mathcal{E}$ *short exact sequences* (or *conflations*).

Chapter 2

Expanding to n -exact categories

After generalizing from abelian categories to exact categories, we can go one step further by looking at longer exact sequences. When not stated otherwise, $\mathcal{A}$ denotes an additive category.

2.1 Working in higher degrees

Before defining n -exact categories, there are many prerequisites that need to be addressed. Firstly, we will need a way to properly describe what the equivalent of a kernel-cokernel pair would be in this new context. As a result, it makes sense to define n -kernels and n -cokernels, which require the notion of weak kernel (respectively weak cokernel).

Definition 2.1 Let $A \xrightarrow{f} B$ be a morphism in $\mathcal{A}$. A *weak kernel* of f is an object $K \in \mathcal{A}$ with a morphism $K \xrightarrow{k} A$ such that the sequence of abelian groups

$$\mathcal{A}(C', K) \xrightarrow{-k} \mathcal{A}(C', A) \xrightarrow{-f} \mathcal{A}(C', B)$$

is exact for all $C' \in \mathcal{A}$.

Dually, a *weak cokernel* of f is an object $C \in \mathcal{A}$ with a morphism $B \xrightarrow{g} C$ such that the sequence of abelian groups

$$\mathcal{A}(C, C') \xrightarrow{k \cdot -} \mathcal{A}(B, C') \xrightarrow{f \cdot -} \mathcal{A}(A, C')$$

is exact for all $C' \in \mathcal{A}$.

Remarks 2.2 We do not ask for the sequence to be short exact, but only for it to be exact in the middle term. Furthermore, the following observations are quite useful in practice:

- (a) Notice the direction of the sequences is different since for weak cokernels, we apply the contravariant functor $\mathcal{A}(-, C')$ to our original sequence, whereas we apply the covariant functor $\mathcal{A}(C', -)$ in the case of weak kernels.
- (b) For $K \xrightarrow{g} A$ a weak kernel of $A \xrightarrow{f} B$, the sequence being exact lets us know that
 - (i) The composition $g \cdot f$ is 0
 - (ii) For any morphism $C \xrightarrow{h} A$ such that $h \cdot f = 0$, there exists a morphism $C \xrightarrow{\hat{h}} K$ such that $\hat{h} \cdot g = h$.

That last point can be summarized by the following diagram:

$$\begin{array}{ccccc}
 K & \xrightarrow{g} & A & \xrightarrow{f} & B \\
 \uparrow \exists \hat{h} & & \nearrow h & \searrow 0 & \\
 C & & & &
 \end{array}$$

These two points follow from exactness in the middle term. Indeed, since $\text{Im} (- \cdot k) = \text{Ker} (- \cdot f)$, any map that, once post-composed with f , gives the 0 map, is in the image of the post-composition with k , i.e. can be factored (not necessarily uniquely!) as $\hat{h} \cdot k$. In particular, if we consider $C' = K$, we conclude that $k = 1_K \cdot k \in \text{Im} (- \cdot k) = \text{Ker} (- \cdot f)$, i.e. $k \cdot f = 0$. The result for weak cokernels is dual.

In summary, weak cokernels and weak kernels have a very similar property to that of cokernels and kernels, except that we lose uniqueness in our factoring of morphisms.

This notion is quite fundamental in the elaboration of n -kernels and n -cokernels:

Definition 2.3 Let $X^0 \xrightarrow{d^0} X^1$ be a morphism in $\mathcal{A}$. An n -kernel of d^0 is a sequence

$$X^{-n} \xrightarrow{d^{-n}} X^{-n+1} \xrightarrow{d^{-n+1}} \dots \xrightarrow{d^0} X^0$$

such that the sequence induced by applying the functor $\mathcal{A}(C', -)$

$$0 \longrightarrow \mathcal{A}(C', X^{-n}) \xrightarrow{- \cdot d^{-n}} \mathcal{A}(C', X^{-n+1}) \xrightarrow{- \cdot d^{-n+1}} \dots \xrightarrow{- \cdot d^0} \mathcal{A}(C', X^1)$$

is exact.

Equivalently, an n -kernel of d^0 is such a sequence $(d^{-n}, d^{-n+1}, \dots, d^{-1})$, where d^{-k} is a weak cokernel of d^{-k+1} for all $1 \leq k \leq n-1$, and d^{-n} is a kernel of d^{-n+1} .

Dually, an n -cokernel of d^0 is a sequence

$$X^1 \xrightarrow{d^1} X^2 \xrightarrow{d^2} \dots \xrightarrow{d^n} X^{n+1}$$

such that d^k is a weak cokernel of d^{k-1} for $1 \leq k \leq n-1$ and d^n is a cokernel of d^{n-1} .

Remark 2.4 In the case $n = 1$, we get back to our usual definition of kernels and cokernels.

Now that we have a suitable notion of higher kernels and cokernels, we can consider the concept of n -exact sequences in the most logical way.

Definition 2.5 An n -exact sequence in $\mathcal{A}$ is a chain complex

$$X^0 \xrightarrow{d^0} X^1 \xrightarrow{d^1} \dots \xrightarrow{d^{n-1}} X^n \xrightarrow{d^n} X^{n+1}$$

such that $(d^0, \dots, d^{n-1})$ is an n -kernel of d^n and $(d^1, \dots, d^n)$ is an n -cokernel of d^0 .

Remark 2.6 Once again, we notice that when $n = 1$, we get back to our usual notion of short exact sequence.

There are two final notions that need to be covered before proceeding with the definition of n -exact categories: that of n -pullback/ n -pushout, and weak isomorphisms. Before covering these concepts, let us first remember what the cone of a morphism of complexes is:

Definition 2.7 Let X and Y be two complexes in $\text{Ch}(\mathcal{A})$ and $f : X \rightarrow Y$ be a morphism of complexes, or diagrammatically:

$$\begin{array}{ccccccc}
 X & & \dots & \longrightarrow & X^{n-1} & \xrightarrow{d_X^{n-1}} & X^n & \xrightarrow{d_X^n} & X^{n+1} & \longrightarrow & \dots \\
 \downarrow f & & & & \downarrow f^{n-1} & & \downarrow f^n & & \downarrow f^{n+1} & & \\
 Y & & \dots & \longrightarrow & Y^{n-1} & \xrightarrow{d_Y^{n-1}} & Y^n & \xrightarrow{d_Y^n} & Y^{n+1} & \longrightarrow & \dots
 \end{array}$$

The *cone* of f , denoted $C(f)$, is the following complex

$$C(f): \quad \dots \longrightarrow X^n \oplus Y^{n-1} \xrightarrow{d_C^n} X^{n+1} \oplus Y^n \longrightarrow \dots$$

where $d_C^n = \begin{pmatrix} -d_X^n & 0 \\ f^n & d_Y^{n-1} \end{pmatrix}$.

Remark 2.8 Let $B^1 \xrightarrow{d_B^1} B^2$ and $A^2 \xrightarrow{f^2} B^2$ be two morphisms in $\mathcal{A}$ and consider the pullback of d_B^1 along f^2 :

$$\begin{array}{ccc}
 A^1 & \xrightarrow{d_A^1} & A^2 \\
 f^1 \downarrow & \text{PB} & \downarrow f^2 \\
 B^1 & \xrightarrow{d_B^1} & B^2
 \end{array}$$

Clearly $f = (f^1, f^2)$ is a morphism of complexes since the square commutes. Thus we can consider its cone:

$$C(f): \quad 0 \longrightarrow A^1 \xrightarrow{d_C^1} A^2 \oplus B^1 \xrightarrow{d_C^2} B^2 \longrightarrow 0$$

where $d_C^1 = \begin{pmatrix} -d_A^1 \\ f^1 \end{pmatrix}$, and $d_C^2 = \begin{pmatrix} 0 & 0 \\ f^2 & d_B^1 \end{pmatrix}$. We claim that d_C^1 is a kernel of d_C^2 . Indeed, consider a morphism $\begin{pmatrix} g \\ h \end{pmatrix}: C \rightarrow A^2 \oplus B^1$ such

that the composition $\begin{pmatrix} g \\ h \end{pmatrix} \cdot d_C^2 = 0$. The composition being 0 implies that $g \cdot f^2 + h \cdot d_B^1 = 0$. Consider the unique map $\hat{h} : C \rightarrow A^1$ given by $\hat{h} \cdot f^1 = h$ and $\hat{h} \cdot d_A^1 = -g$ (the map is given by the pullback property; the fact that $-g \cdot f^2 = h \cdot d_B^1$ comes from the previous remark about the composition being 0). We see that $\hat{h} d_C^1 = \begin{pmatrix} -\hat{h} \cdot d_A^1 \\ \hat{h} \cdot f^1 \end{pmatrix} = \begin{pmatrix} g \\ h \end{pmatrix}$. Thus we have defined a unique map $\hat{h} : C \rightarrow A^1$ such that $\hat{h} \cdot d_C^1 = \begin{pmatrix} g \\ h \end{pmatrix}$. That is, d_C^1 is a kernel of d_C^2 .

A dual result is true for pushouts, where d_C^2 is a cokernel of d_C^1 .

Basing ourselves on this result, we can generalize the notion of pullback and pushout to higher degrees:

Definition 2.9 Let $X : X^0 \xrightarrow{d_X^0} X^1 \xrightarrow{d_X^1} \dots \xrightarrow{d_X^{n-1}} X^n$ be a complex in $\text{Ch}(\mathcal{A})$ and $f^0 : X^0 \rightarrow Y^0$ a morphism in $\mathcal{A}$. An n -pushout of X along f^0 is a morphism of complexes

$$\begin{array}{ccccccc} X : & X^0 & \xrightarrow{d_X^0} & X^1 & \xrightarrow{d_X^1} & \dots & \xrightarrow{d_X^{n-1}} & X^n \\ & \downarrow f & & \downarrow f^0 & & \downarrow f^1 & & \downarrow f^n \\ Y : & Y^0 & \xrightarrow{d_Y^0} & Y^1 & \xrightarrow{d_Y^1} & \dots & \xrightarrow{d_Y^{n-1}} & Y^n \end{array}$$

such that in the cone of f :

$$C(f) : \quad X^0 \xrightarrow{d_C^{-1}} X^1 \oplus Y^0 \xrightarrow{d_C^0} \dots \xrightarrow{d_C^{n-1}} X^n \oplus Y^{n-1} \xrightarrow{d_C^n} Y^n$$

$(d_C^0, \dots, d_C^n)$ is an n -cokernel of d_C^{-1} .

The definition for the n -pullback of a chain complex is dual. That is, given a complex Y in $\text{Ch}^n(\mathcal{A})$ and a morphism $f^n : X^n \rightarrow Y^n$, an n -pullback of Y along f^n is a morphism of complexes $f : X \rightarrow Y$ such

that in its cone $C = C(f)$, the sequence $(d_C^{-1}, \dots, d_C^{n-1})$ is an n -kernel of d_C^n .

The last notion required for defining n -exact categories is that of weak isomorphisms:

Definition 2.10 Let X and Y be two n -exact sequences in $\mathcal{A}$. We say $f : X \rightarrow Y$ is a *weak isomorphism* of complexes if there is $k \in \{0, 1, \dots, n\}$ such that f^k and f^{k+1} are isomorphisms.

When $n = 1$, we note that weak isomorphisms are isomorphisms of short exact sequences, though the result is not immediate. For the sake of conciseness, we will not give a proof of this result. In the general case, however, weak isomorphisms are not necessarily isomorphisms, but a weaker result still holds:

Proposition 2.11 *Let X and Y be two n -exact sequences in $\mathcal{A}$ and let $f : X \rightarrow Y$ be a morphism of complexes. If f^k and f^{k+1} are isomorphisms for some $k \in \{0, 1, \dots, n\}$, then f induces an isomorphism in $H(\mathcal{A})$.*

The proof of this result can be found in [2, Chapter 2, Proposition 2.7].

All of this preliminary work being done, we can finally proceed to define n -exact categories.

2.2 n -exact categories

With the proper tools put in place, it is quite straightforward to expand the definition of exact categories to longer exact sequences. Similar to what was done in chapter 1, we proceed with admissible n -exact sequences, monics, and epics:

Definition 2.12 Let $\mathcal{E}$ be a class of n -exact sequences in $\mathcal{A}$. An $\mathcal{E}$ -admissible n -exact sequence is an n -exact sequence that is in $\mathcal{E}$. An $\mathcal{E}$ -admissible monic is a morphism $d_X^0 : X^0 \rightarrow X^1$ such that there is an $\mathcal{E}$ -admissible n -exact sequence $X^0 \xrightarrow{d_X^0} X^1 \xrightarrow{d_X^1} \dots \xrightarrow{d_X^n} X^{n+1}$. Dually, an $\mathcal{E}$ -admissible epic is a morphism $d_X^n : X^n \rightarrow X^{n+1}$ such that there is an $\mathcal{E}$ -admissible n -exact sequence $X^0 \xrightarrow{d_X^0} \dots \xrightarrow{d_X^{n-1}} X^n \xrightarrow{d_X^n} X^{n+1}$.

Whenever the context is clear, we will simply say admissible n -exact sequence (respectively admissible monic, admissible epic). We shall write $X^0 \xrightarrow{d_X^0} X^1$ for admissible monics and $X^n \xrightarrow{d_X^n} X^{n+1}$ for admissible epics.

The following definition simply adapts the axioms of exact categories to the definitions given in the previous section:

Definition 2.13 Let $\mathcal{E}$ be a class of n -exact sequences on $\mathcal{A}$. $\mathcal{E}$ is said to be an n -exact structure on $\mathcal{A}$ if it is closed under weak isomorphisms and it satisfies the following axioms:

- [E0] The sequence $0 \rightarrow 0 \rightarrow \dots \rightarrow 0$ is an admissible n -exact sequence;
- [E1] The composition of two admissible monics yields an admissible monic;
- [E1^{op}] The composition of two admissible epics yields an admissible epic;
- [E2] If $X : X^0 \xrightarrow{d_X^0} \dots \xrightarrow{d_X^n} X^{n+1}$ is an admissible n -exact sequence and $X^0 \xrightarrow{f^0} Y^0$ is a morphism in $\mathcal{A}$, there exists an n -pushout of $(d_X^0, \dots, d_X^{n-1})$ along f^0

$$\begin{array}{ccccccc}
 X : & X^0 & \xrightarrow{d_X^0} & X^1 & \xrightarrow{d_X^1} & \dots & \xrightarrow{d_X^{n-1}} X^n \xrightarrow{\dots d_X^n} X^{n+1} \\
 \downarrow f & \downarrow f^0 & & \downarrow f^1 & & & \downarrow f^n \\
 Y : & Y^0 & \xrightarrow{d_Y^0} & Y^1 & \xrightarrow{d_Y^1} & \dots & \xrightarrow{d_Y^{n-1}} Y^n
 \end{array}$$

such that d_Y^0 is an admissible monic;

[E2^{op}] If $Y : Y^0 \xrightarrow{d_Y^0} \dots \xrightarrow{d_Y^n} Y^{n+1}$ is an admissible n -exact sequence and $X^{n+1} \xrightarrow{f^{n+1}} Y^{n+1}$ is a morphism in $\mathcal{A}$, there is an n -pullback of $(d_Y^1, \dots, d_Y^n)$ along f^{n+1}

$$\begin{array}{ccccccc}
 X : & & X^1 & \xrightarrow{d_X^1} & \dots & \xrightarrow{d_X^{n-1}} & X^n & \xrightarrow{d_X^n} & X^{n+1} \\
 \downarrow f & & \downarrow f^1 & & & & \downarrow f^n & & \downarrow f^{n+1} \\
 Y : & Y^0 & \xrightarrow{d_Y^0} & Y^1 & \xrightarrow{d_Y^1} & \dots & \xrightarrow{d_Y^{n-1}} & Y^n & \xrightarrow{d_Y^n} Y^{n+1}
 \end{array}$$

such that d_X^n is an admissible epic.

Chapter 3

Generalizing results from
exact to n -exact categories

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